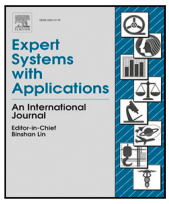
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# Learning face super-resolution through identity features and distilling facial prior knowledge

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ABSTRACT

Deep learning techniques in electronic surveillance have shown impressive performance for super-resolution (SR) of captured low-quality face images. Most of these techniques adopt facial priors to improve the feature details in the resultant super-resolved images. However, the estimation of facial priors from the captured low-quality images is often inaccurate in real-life situations because of their tiny, noisy, and blurry nature. Thus, the fusion of such priors badly affects the performance of these models. Therefore, this work presents a teacher–student-based face SR framework that efficiently preserves the personal facial structure information in the super-resolved faces. In the proposed framework, the teacher network exploits the facial heatmap-based ground-truth-prior to learn the facial structure that is utilized by the student network. The student network is trained with the identity feature loss for maintaining the identity and facial structure information in reconstructed high-resolution (HR) face images. The performance of the proposed framework is evaluated by conducting the experimental study on standard datasets namely CelebA-HQ and LFW face. The experimental results reveal that the proposed technique conquers the existing methods for the face SR task.

## 1. Introduction

Face super-resolution (SR) refers to a domain-specific SR problem where the task is to infer the high-resolution (HR) version from the given degraded low-resolution (LR) observation. Due to the constraints of consumer imaging devices, the obtained real-world images often get blurred, noisy, and spatially small. These images do not carry adequate information, and create problems in numerous applications (Aribilola, Asghar, Kanwal, Fleury, & Lee, 2022; Du et al., 2022; Ghosh, Vatsa, & Singh, 2022; Li & Cheung, 2021; Mehta & Murala, 2022; Qu et al., 2021; Wen et al., 2019) such as face recognition, facial expression analysis, intelligent surveillance, and so on. The performance of these tasks suffered severely due to the LR version of the face images. Therefore, face SR has become essential for enhancing the accuracy of face-related vision tasks.

The face SR is an innately ill-posed problem because there may be numerous likely HR versions corresponding to the given LR face image. As the face SR task has an extremely large solution space, therefore, taking advantage of informative facial priors becomes necessary to generate a unique HR image.

Nowadays, Convolutional Neural Networks (CNN) have been widely applied in different areas of computer vision, such as classification (Tu, Zhou, Kuang, Chen, & Plaza, 2022), image captioning (Ak, Sun, & Lim, 2022), health-related applications (Iwamoto, Takeda, Li, Shiino, & Chen, 2022), blind image SR (Li et al., 2022; Wu et al., 2022), and image reconstruction (Qin, Bai, & Zhao, 2022; Tomar, Arya, & Rajput, 2023a). In this direction, SR methods (Jiang, Wang, Yi, Wang, Lu, et al., 2019; Xiao et al., 2021, 2023) have been developed using the GAN framework and deformable convolution to enhance the satellite and remote sensing images. Firstly, the network is trained by applying the loss function to perform the specific task. After that, the trained network is used to predict the outcome from the given test input. Similarly, the SR of the face images took place by attempting to learn the relationship among the LR and corresponding HR pixels in the dataset, and utilizing the same during the test phase to generate the HR version of the test LR images. Generative Adversarial Networks (GAN) have shown the powerful ability to generate realistic images from random noise, and significantly contributed in the areas of computer vision, such as style transfer (Huo et al., 2022), inpainting (Jain, Zhou, Yu, & Shi, 2023), and super-resolution of the images (Marenzi, Torti, Leporati, Quevedo, & Callicó, 2017). In the GAN-based framework, two neural networks contest a game named as zero-sum to maximize and minimize their loss functions. In such a way, these networks are trained

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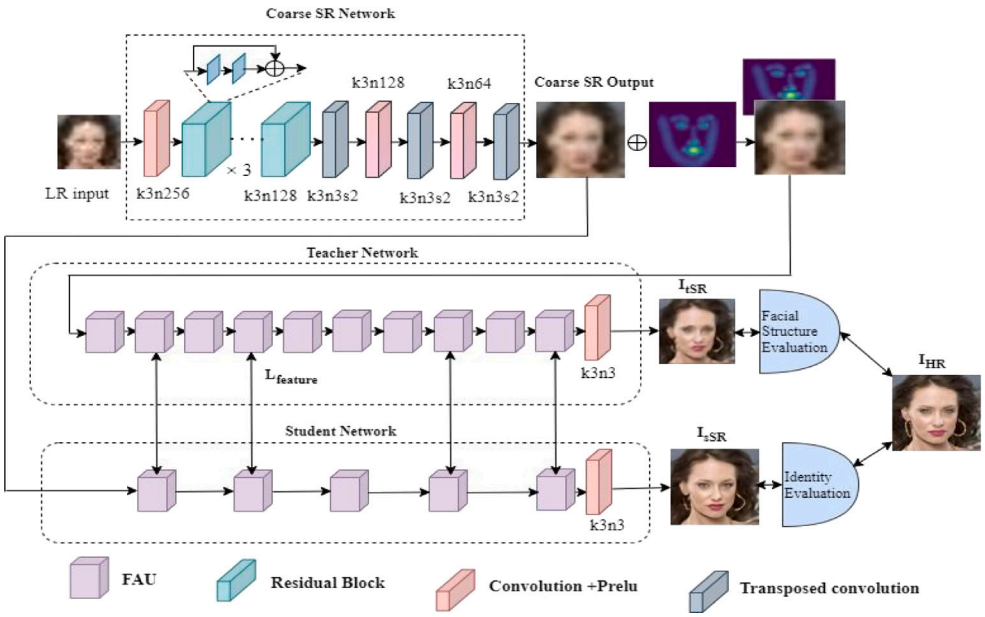



**Fig. 1.** An working overview of the proposed knowledge distillation based face super-resolution framework.

adversarially, which helps not only to produce visually plausible images but also generate texture-preserving HR images.

As Facial images possess the structural details for visual perception and also carry identity information. Hence, it is worthwhile to utilize specific facial details i.e., facial structure and identity information during the face SR process. It plays an essential role in comprehending the structural features and localizing the facial components of human faces. The facial landmarks are used to generate the component heatmap that provides information related to the spatial location of components in the face images. Similarly, the identity features constraint helps the SR process to enforce the identity information in the output faces.

In previous works (Bulat & Tzimiropoulos, 2018; Chen, Tai, Liu, Shen, & Yang, 2018; Wang, Jiang, Zhong, Zhai & Liu, 2023; Yu, Fernando, Ghanem, Porikli, & Hartley, 2018), many SR methods have been developed by incorporating the facial prior in SR network. Most of them estimate the prior from the LR image and concatenate with the input image or intermediate feature maps to propagate the structure-related information in the successive layers of the model. However, it is hard to predict accurate facial structure information due to the limited details available in the consumer's gadgets captured noisy LR images. Afterward, researchers (Kim, Kim, Kwon, & Kim, 2019; Ma, Jiang, Rao, Lu, & Zhou, 2020) moved to extract the facial prior from the coarse images (upsample LR images) or intermediate super-resolved images which significantly improved the performance of the face SR methods. As during initial training iteration these images do not carry adequate facial components, hence, the estimation of accurate prior is still challenging from these images. The incorporation of such inaccurate prior may amplify the error in subsequent iterations, causing deterioration of the output faces.

To address the above shortcomings of the existing techniques (Chen, Gong, Wang, Li, & Wong, 2020; Chen et al., 2018; Dastmalchi & Aghaeinia, 2022; Ma et al., 2020), we propose a teacher–student-based face SR framework (shown in Fig. 1) that explores facial heatmap-based ground truth prior with identity constraints at the training stage. The LR image is only taken as input during the inference phase to overcome the issue of estimating accurately prior. Here, the teacher network aims to learn the prior information which is propagated to the student network through a distillation strategy. Further, the student network (trained with only the pair of LR and HR images) constrains the generated images in the identity domain which facilitates the network to preserve the identity more accurately. The proposed framework requires only the LR images as input in the test phase to generate the HR images. In this way, the proposed framework avoids the issue of estimating the accurate prior during the test phase and restores the facial structure and identity details in the output face images.

The significant contributions of this research work are as follows:

- Facial Heatmap-based Ground Truth Prior is used with a knowledge distillation approach to get more appropriate features for the face SR process.
- The feature attention unit (FAU) consists of the feature and the attention block is designed to learn the micro and high-level facial details.
- Multiple FAUs are applied to continuously enhance the features and train the network with an assemble loss function to hold the identity and structure details in the resultant images.

The remaining fragments of this research article are organized as follows. Section 2 presents a detailed review of the face super-resolution. The proposed methodology and loss function are elaborated in Section 3. The details of the experimental setup, implementation details, and results of the experimental outcome are discussed in Section 4. Finally, the conclusions of this research are presented in Section 5.

## 2. Literature review

The pioneering work of Baker and Kanade (2000) demonstrated the face SR method to enhance the resolution of the input LR images. In this, similar LR images were explored from the dataset for a given input to discover the underlying mapping between them. These learned mappings are applied to infer the corresponding HR image. Since then, the face SR has attained more attraction in the area of computer vision which inspired researchers to develop better approaches for further improving the quality of the output HR face images. A technique designed by Liu, Shum, and Zhang (2001) applied global and local-based learning to improve SR performance. The global model recovered the coarse image, and high-frequency details are restored by the patch-based approach. The Eigen transformation-based approach was developed by Wang and Tang (2005), where input image is projected over the eigenfaces to compute the representation coefficient. Further, the incorporation of regularization (Gao et al., 2021; Rai & Rajput, 2023, 2024; Rajput, 2022; Wang, Hu, Wang, & Jiang, 2013) has

improved the stability and robustness of face SR. A weighted adaptive sparse method (Wang et al., 2013) was designed to take benefits of adaptive sparse that helps to learn the specific sparsity corresponding to the observed face images. After that, local structure prior (Gao et al., 2021), iterative sparsity (Rajput, 2022), and many other regularization-based methods (Jeong, Yoon, & Paik, 2015; Kumar & Rajput, 2023; Rai & Rajput, 2022; Rajput, 2023; Rajput & Arya, 2019; Rajput, Arya, & Singh, 2018; Rajput, Rai, & Kumar, 2024; Rajput, Singh, Arya, & Jiang, 2018) were exploited to boost the performance of face SR. The above-discussed SR approaches primarily introduced a variety of regularization terms and optimize the objective function to learn the representation coefficients. However, due to limited representation capability such techniques fail to restore the face image details for large-scale factors.

To address these problems, deep learning-based methods have been designed which have shown outstanding learning abilities from the input images. In this way, they learn better details which makes them superior then traditional methods. The first CNN-based network was designed by Dong, Loy, He, and Tang (2015) for the task of image SR which motivated the researcher to apply that for face images. After that, a novel Bi-channel CNN architecture (Zhou, Fan, Cao, Jiang, & Yin, 2015) was proposed firstly for the face SR. It extracts the features to generate the intermediate output which is fused with the upsampled LR image to reconstruct the output HR images. Further, the discovery of the GAN (Goodfellow et al., 2014) and its ability to generate images gave a new direction to face SR methods. Later many GAN-based frameworks are proposed which used the adversarial loss to hold the texture details in the generated HR face images. Although, these are powerful networks but their nature of considering all features equally important may cause the introduction of artifacts in the reconstruction process, especially for noisy LR input images. To overcome this issue, Chen et al. (2020) proposed a framework that used residual attention blocks for feature extraction which allow the CNN layers to pay more attention to informative features. It helps this model to minimize the effect of artifacts from the resultant face images. An ensemble learning approach has been used in Jiang, Wang, Yi, Wang, Gu, et al. (2019) to combine multiple model's outputs. It helps to exploit the multiple models' benefits, and thus, such fusion assists in preserving informative details in resultant images. Jiang et al. (2020) designed a network consisting of multiple branches responsible for learning the global and local details. Finally, these pieces of information were fused to generate images with better texture. The novel method presented in Lu et al. (2021) splits the input features and then applied attention on feature-rich regions. The work of Chen et al. (2020) has been extended in Tomar, Arya, and Rajput (2023b, 2023c) which use multi-scale features with attention to improving the face SR performance. After that, a dual-branch network-based face SR method (Zengy et al., 2024) was proposed where two networks interact and learn the local texture and global facial structure details to improve the overall quality of generated faces. However, such methods may lose a few beneficial information which cause to loss of facial details in generated images. Therefore, the researcher developed the prior guided SR methods which utilized the facial prior knowledge while super-resolving the input images. The first prior guided face SR method was developed by Chen et al. (2018) where they embed the facial geometry prior with CNN layers to generate visually promising results. Yu et al. (2018) fused the extracted features with facial component maps to guide the SR network. It assists in recovering the facial structure details in the generated images. A network (Wang, Jiang, Jiang, & Liu, 2024) was designed to leverage facial structure prior to perform the SR task and preserve the texture and facial contour details in resultant face images. Although, these methods highly relied on accurate prior information to improve their performance. Thus, incorrect estimated prior may cause the introduction of erroneous facial knowledge which may be expanded iteratively during the training of the model. To overcome this, a new approach was proposed in Wang, Jiang, Zhong, and Liu (2022) that utilized a ground-truth facial parse map for better recovery of sharp edges. Inspired by this research, we propose a face SR framework to generate identity and structure-preserving face images. However, the proposed approach is different from Wang et al. (2022) in several ways: (1) fuses the output feature maps of the multi-scale filters to realize a better representation of the input; (2) employs the attention mechanism to enhance the informative features and suppress the poor-quality features; (3) regularize the output in identity and landmark space to preserve the facial structure in the generated HR images.

## 3. The proposed work

Recently, by utilizing the facial prior, deep learning-based face SR methods (Chen et al., 2021, 2018; Ma et al., 2020; Yu, Zhang, & Xie, 2021) have demonstrated significant performance improvement over conventional face SR methods. These methods integrated the different facial priors to supervise the SR process. However, such techniques have the following drawbacks: (i) they often focused on generating visually promising images without holding the identity features. (ii) Estimation of accurate facial prior directly from the LR face image is challenging. Consequently, propagation of inaccurate prior may cause the generation of an image with wrong facial details. (iii) Facial prior is employed with the LR image, which is too indistinct and does not carry the coarse features. Consequently, it may fail to be effectively utilized prior in the training phase and causes the model to generate a distorted output image. (iv) Use of the fixed-size convolutional filters restricts these models to concentrate only on a specified receptive field to detect the characteristics which results in the ignorance of the spatial structure details. (v) Assigning the same significance to features may cause noisy and irrelevant feature propagation in the reconstruction process of these models, leading to poor-quality output faces.

Thus, a method is required which can incorporate facial prior effectively to generate HR images with better facial structure and identity-preservation. Therefore, we propose a framework that learns identity features and distilling facial prior knowledge for face super-resolution.

The proposed framework, illustrated in Fig. 1, consists of the two modules: (i) Coarse SR network, and (ii) Teacher and student networks. Instead of traditional bicubic interpolation used in the existing state-of-the-art works (Chen et al., 2020; Dong, Loy, He, & Tang, 2014; Kim, Lee, & Lee, 2016; Wang et al., 2022), a coarse SR network is designed in this work to upsample the LR images. It helps the proposed framework in recovering the coarse features through model training. The teacher network (TN) (trained with a pair of concatenated output of coarse network with prior and HR images) utilizes the facial heatmap-based ground truth prior (FHGTP), while the student network (SN) is trained only with the pair of LR and HR face images. The proposed framework first upsamples the input images and then applies the high-quality FHGTP with them to utilize prior effectively in the TN. Further, we designed the student network that takes the benefits of facial structure knowledge and attentive features through the knowledge distillation approach. Moreover, in contrast to the existing methods, this work explored the multi-scale filters with attention to extract the informative features. This knowledge of features is transferred from the TN to the SN through a feature distillation approach. The proposed framework only uses the coarse network and SN in the inference phase which takes the LR image as input. The FHGTP is unavailable during the inference phase, which the SN highly requires. During training, the TN transfers the FHGTP knowledge to the SN which assists in reconstructing the HR images in the inference phase. The convolutional kernel, stride size, and the number of filters used in the CNN layer are represented as k3s2n128 which denotes the kernel size $3 \times 3$ with stride 2, and 128 such filters. The details of the Coarse network, teacher, and student network are discussed below.

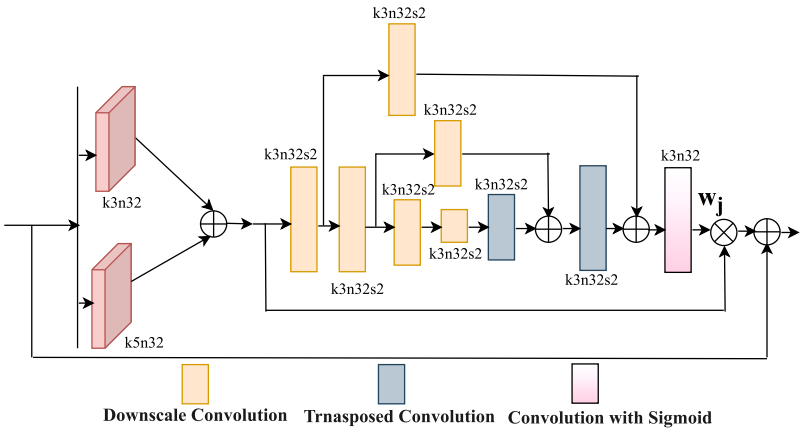



**Fig. 2.** Feature attention unit (FAU).

### 3.1. Coarse SR network

It is employed to recover the facial features in the coarse image. The popular residual blocks are used for extracting the features from the input image which is followed by the transposed convolutional layers to upsample the features. After that, the convolutional layer with linear activation is applied to recover the upsampled LR image with coarse features. The mathematical formulation of the coarse SR module is defined by Eq. (1).

$$I_c = CN_{\theta_c}(I_{LR}) \tag{1}$$

where $CN$ and $I_c$ denote the coarse network and its output (i.e., coarse SR image) respectively ; while $\theta_c$ implies the learned parameters of the network. This output is concatenated with the facial structure prior which is given as the input to the TN.

### 3.2. Teacher network (TN)

In the proposed framework the TN is designed to exploit the FHGTP that provides information about the spatial location of landmarks and helps to boost the face SR performance. In contrast to existing methods (Chen et al., 2020; Wang et al., 2022), the proposed framework first generates the coarse SR version of the input LR image and then combines it with the FHGTP to overcome the effect of artifacts from SR process. The detailed architecture of the TN is discussed below.

#### 3.2.1. Architecture

The TN comprises multiple Feature Attention Units (FAU) which extract and utilize the informative deep features to generate images. It consists of a multi-scale filter-based Feature block and a spatial Attention block to assign a higher weight to informative features as illustrated in Fig. 2. These FAUs are repeated continuously to extract high-quality features, which is formulated in Eq. (2).

$$x_n = f^n_{FAU}\left(f^{n-1}_{FAU}\left(\ldots f^0_{FAU}\left([I_c + FHGTP]\right)\right)\right) \tag{2}$$

where $f^n_{FAU}$ and $x_n$ denote the $n$th FAU and its output respectively; while $[I_c + FHGTP]$ represents the concatenation of FHGTP with coarse SR image. Finally, the convolutional layer with filter size 3 × 3 is applied to generate the output images ($I_{tSR}$).

FAU applies different size filters to extract the facial features, but not all of these features may be beneficial for SR. Hence, assigning the same priority to such features may degrades the quality of the output face image. Therefore, attention is applied to more focus on the informative facial features. For this purpose, we use a spatial attention map that generates the weights for each spatial location of the feature maps. It helps in assigning the weights to features as per their importance. Instead of applying a fixed size filter that may fail to analyze the different pattern sizes features, the proposed feature block applies the two different size filters (3 × 3 and 5 × 5) which use different receptive fields to detect the features. Finally, the output of these filters is fused to hold the micro to high-level details of features. It helps the proposed framework to persist the different variations of details while recovering the face images. The mathematical formulation of the feature block is represented by Eq. (3).

$$f(y) = \mu_{conv/3\times3}(y) + \mu_{conv/5\times5}(y) \tag{3}$$

where the convolution operation with kernel size k over the input $y$ is denoted by $\mu_{conv/k\times k}(y)$ and $f(y)$ describes the output of the feature block.

The FAU integrates the feature block with the Attention unit to refine the features by considering the holistic appearance of the image. In this way, the model can effectively prioritize informative details even in noisy input images. Additionally, it focuses on informative facial features by integrating the multiple scales features to capture diverse information about facial details and enable the learning of the comprehensive representation of the input. The attention unit dynamically adjusts weights to improve the model's robustness against the noisy input. Each FAU block enables the network to diminish the noise and enriches the feature details. Further, the attentive features are integrated with the outcome of the last layer to produce the fused output which holds advantageous features. In this way, the outcome of each block is generated and forwarded to successive FAU to refine the features.

#### 3.2.2. Loss function

The proposed TN applies a loss function to learn their parameters for regularizing the output in specific spaces. For this purpose, a assemble loss is designed to maintain the output consistency in the pixel and facial structure space. The assemble loss function consists of the following components.

1. Pixel Loss: The $L_1$ norm between the pixels of HR and output images is applied to calculate the pixel loss. It assists in the recovery of the low-level content in the generated images which is computed as:

$$L_{pix} = \frac{1}{N}\sum_{i=1}^{N} \|I^i_{tSR} - I^i_{HR}\|_1 \tag{4}$$

where $L_{pix}$ and $N$ denote the loss function based on pixel-wise and the number of training images, respectively; $I_{tSR}$ and $I_{HR}$ represent the TN output image and HR image, respectively.

2. Heatmap Loss: Heatmap is generated by drawing the facial boundary corresponding to facial landmarks as shown in Fig. 3. For this, we estimate the coordinates of the facial landmarks by providing the Ground Truth images as input to face alignment




**Fig. 3.** Process of generating the heatmap. (a) HR image, (b) facial landmarks, (c) facial landmarks based heatmap of HR images also known as FHGTP, and (d) face images superimposed heatmap.

algorithm (Bulat & Tzimiropoulos, 2017). The landmark's coordinates are set as mean while computing the Gaussian distance between landmarks points and other pixels of image. Hence, the intensity becomes highest on the boundary. In this way, heatmap is generated for any input image. Based on that, heatmap loss is computed which helps to constrain the facial landmarks in the generated images. It is represented as:

$$\mathcal{L}_h = \frac{1}{N}\sum_{i=1}^{N}\left\|h_s^i - h_{hr}^i\right\|_1 \tag{5}$$

where $h_s^i$ and $h_{hr}^i$ represent the facial boundary heatmap of $i$th TN output image and Ground Truth image, respectively.

Finally, the assemble loss function ($L_{AT}$) for the teacher network is designed as follows.

$$L_{At} = \lambda_{pix} L_{pix} + \lambda_h L_h, \tag{6}$$

where $\lambda_{pix}$ and $\lambda_h$ represent weights of pixel and heatmap loss, respectively.

### 3.3. Student network (SN)

The student network takes benefit of FHGTP to distill the prior information from TN. It assists the network in grabbing the facial prior knowledge that is applied to reconstruct HR images. In this way, the SN avoids the propagation of inaccurate estimated prior. Moreover, the identity feature-based loss function is used to train the SN which helps the network in preserving the identity features in the output images. The details of the SN is discussed below.

#### 3.3.1. Architecture

The SN comprises of FAUs, and follows a similar architecture to the TN. In contrast to the TN, it does not take the FHGTP as input but utilizes the prior knowledge via distillation loss to super-resolve the input images. In addition, assemble loss function consisting of pixel loss, identity feature loss, and distillation loss to train the SN. It helps the network to recover the identity and structure-preserving face images from the input LR images during the inference phase.

#### 3.3.2. Loss function

The assemble loss function of the SN contains the following components:

1. Pixel Loss: It is computed as the pixel-wise difference between the super-resolved images of the SN and Ground Truth images. Mathematically, it is denoted as:

$$L_{pix} = \frac{1}{N}\sum_{i=1}^{N}\|I_{sSR}^i - I_{HR}^i\|_1 \tag{7}$$

where $I_{sSR}$ and $I_{HR}$ represent the output of the SN and Ground Truth image, respectively.

2. Identity Loss: The pre-trained VGG-based face recognition model is applied over the Ground Truth and generated images of SN to represent their identity information in terms of intermediate layer's features. Further, the identity loss is computed as the absolute difference between these features which is represented in Eq. (8).

$$\mathcal{L}_{id} = \frac{1}{N}\sum_{i=1}^{N}\left\|f_{id}\left(I_{sSR}^i\right) - f_{id}\left(I_{HR}^i\right)\right\|_1 \tag{8}$$

where $f_{id}$ represents the output of the pre-trained face recognition model.

3. Distillation Loss: The distillation loss is employed to utilize the facial prior information in the SN. To accomplish this, we regularize the SN's output and their features at intermediate layer in the domain of the TN. It is composed of the following components.

a Soft Ground Truth Loss: The soft ground truth loss ($L_{sg}$) is computed as the pixel-wise intensity difference between the output image of SN and the resultant image of TN (known as soft ground truth), which is formulated as:-

$$L_{sg} = \frac{1}{N}\sum_{i=1}^{N}\left\|I_{sSR}^i - I_{tSR}^i\right\|_1 \tag{9}$$

where $I_{sSR}$ denotes the output image of the SN.

b Feature Space Loss: The individual training of both models (i.e., TN and SN) with different loss functions entitles their features to learn the specific knowledge. As SN requires informative attributes of the facial prior to recover the fine texture during the reconstruction process. Therefore, feature similarity loss is applied to acquire prior knowledge. It is calculated as a feature-wise difference between the intermediate layers of both models

which is defined as:

$$\mathcal{L}_{\text{Feature}} = \sum_{i}^{n} \frac{1}{n} \left\| f_{\text{s}}^{\text{i}} - f_{\text{t}}^{\text{i}} \right\|_1 \tag{10}$$

where $f_t^i$ and $f_s^i$ represent $i^{\text{th}}$ layer features of TN and SN, respectively.

The assemble loss function is designed to train the SN which is presented in Eq. (11)

$$L_{As} = \lambda_{pix} L_{pix} + \lambda_{id} L_{id} + \lambda_{sg} L_{sg} + \lambda_f L_{Feature}, \tag{11}$$

where $\lambda_{pix}$, $\lambda_{id}$, and $\lambda_{sg}$ are weights of pixel, identity loss, and distillation loss, respectively.

## 4. Performance evaluation

This section gives the details about the datasets, experimentation setup, results, and performance analysis.

### 4.1. Datasets

The two-face datasets namely CelebA-HQ (Karras, Aila, Laine, & Lehtinen, 2018) and LFW (Learned-Miller, Huang, RoyChowdhury, Li, & Hua, 2016) are used for experiments in this research which comprise of 30 K and 13 K face images, respectively. These images constitute background information and are inconsistent in the terms of their dimensions. Therefore, the Multitask cascaded CNN (MTCNN) (Zhang, Zhang, Li, & Qiao, 2016) is applied to detect the facial regions and crop out the face images. These images are resized and adjusted to $128 \times 128$ pixels which act as the HR images. The down-sample operation over the HR images are performed to get the LR images of size $32 \times 32$ pixels, $16 \times 16$ pixels. After that, Gaussian noise with the standard deviation of 10/255 is introduced in the LR images. These images are taken as input by the model to produce the output images of size $128 \times 128$ pixels with upscale factors 4 ×, and 8 ×. For training, 28 K and 12 K images and for testing 2 K and 1 K images from the CelebA-HQ and LFW respectively are used.

The peak signal-to-noise ratio (PSNR), structure similarity (SSIM) (Wang, Bovik, Sheikh, & Simoncelli, 2004), and learned perceptual image patch similarity (LPIPS) (Zhang, Isola, Efros, Shechtman, & Wang, 2018) metrics are considered to assess the quality of generated face images. As super-resolved faces are widely used for human identification in numerous real-world applications, performance of the proposed framework and other compared methods are also analyzed based on identity similarity and preservation of the landmarks in the face images.

### 4.2. Implementation details

Firstly, the TN is trained with $L_{At}$ loss and frozen their parameters. After that, training of the SN is done based on distillation loss. Then, facial identity feature loss is applied to constrain the output images in the identity space. Further, both networks are optimized using *Adam* with a fixed learning rate of $10^{-4}$ and batch size two. The parameters used in the assemble loss function of TN are set as: $\lambda_{pix} = 100$, and $\lambda_h = 1$; while for SN $\lambda_{pix} = 100$, $\lambda_{id} = 1$, $\lambda_f = 0.1$, and $\lambda_{sg} = 10$.

### 4.3. Results

The obtained results are compared with different SR models namely SRCNN (Dong et al., 2014), VDSR (Kim et al., 2016), SRGAN (Ledig et al., 2017), URDGN (Yu & Porikli, 2016), SISN (Lu et al., 2021), FSRNet (Chen et al., 2018), DICNet (Ma et al., 2020), WIPA (Dastmalchi & Aghaeinia, 2022), HyFeat (Tomar et al., 2023c), SPARNet (Chen et al., 2020), SFMNet (Wang, Jiang, Zhong & Liu, 2023), and CTCNet (Gao et al., 2023) to demonstrate the effectiveness of the proposed work. The quantitative and qualitative analysis of results are performed to assess the performance of these methods.

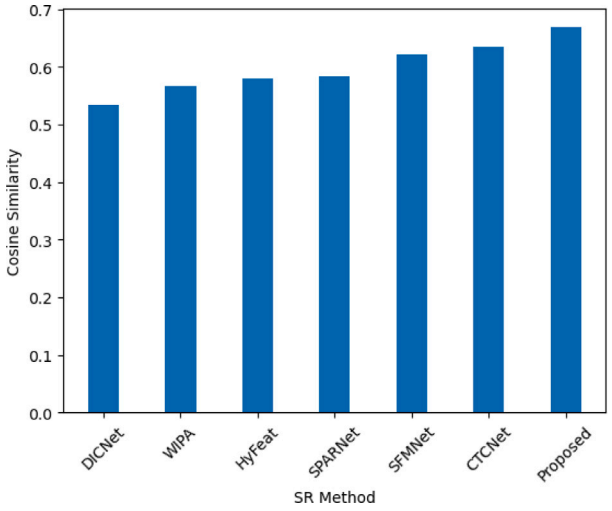



**Fig. 4.** Comparison of different SR methods in terms of average cosine similarity.

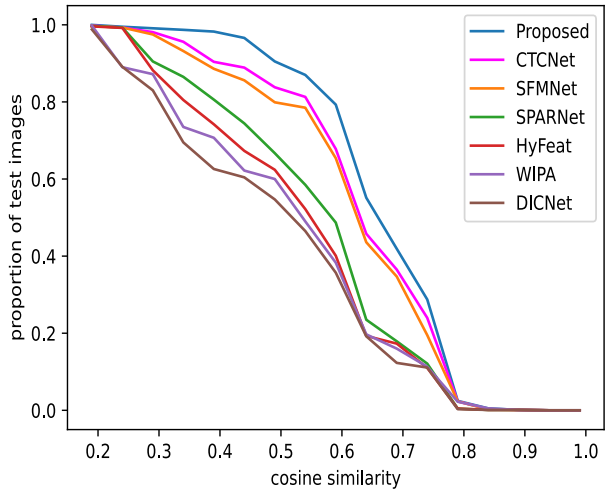



**Fig. 5.** Investigation of the identity information recovery in the output images by different SR methods.

#### 4.3.1. Quantitative comparisons

The quantitative comparison of results is demonstrated by values of PSNR, SSIM, and identity similarity metrics. Table 1 shows the results[1] of 4 × and 8 × upsampling tasks by distinct SR methods for two different datasets. It indicates the outstanding performance of the proposed framework over the existing methods due to distilling facial prior knowledge and learning the identity features. In general, for higher upscale factor (i.e., > 4 ×), the super-resolved images fail to hold the identity features which makes identification task challenging. Therefore, we adopted identity similarity metrics to compare the outcome of the proposed framework with recent face SR methods (Chen et al., 2020; Dastmalchi & Aghaeinia, 2022; Gao et al., 2023; Ma et al., 2020; Tomar et al., 2023c; Wang, Jiang, Zhong & Liu, 2023). To compute this, a pretrained face recognition model (Serengil & Ozpinar, 2020) is utilized that takes the images as input to fetch their identity feature vector. These vectors are used to calculate the cosine similarity that indicates the likeliness of carrying the identity information between face images. The computed identity similarity results are depicted in

[1] As pretrained models of Dastmalchi and Aghaeinia (2022), Gao et al. (2023), Ma et al. (2020) work with 8 × upscale factor. Hence, their results are considered for only 8 × scale.

**Table 1**
Quantitative assessment of the various SR methods for 4 × and 8 × scale. The bold and underlined values show the best and second-best scores, respectively.

| Model | Scale | Dataset | | | | | |
|---|---|---|---|---|---|---|---|
| | | LFW | | | CelebA-HQ | | |
| | | PSNR | SSIM | LPIPS | PSNR | SSIM | LPIPS |
| SRCNN (Dong et al., 2014) | 4 × | 27.75 | 0.670 | 0.349 | 27.78 | 0.710 | 0.327 |
| VDSR (Kim et al., 2016) | 4 × | 27.86 | 0.720 | 0.279 | 27.97 | 0.743 | 0.273 |
| SRGAN (Ledig et al., 2017) | 4 × | 28.23 | 0.748 | 0.234 | 28.65 | 0.758 | 0.232 |
| URDGN (Yu & Porikli, 2016) | 4 × | 28.14 | 0.742 | 0.203 | 28.57 | 0.742 | 0.196 |
| SISN (Lu et al., 2021) | 4 × | 28.46 | 0.756 | 0.184 | 28.79 | 0.764 | 0.182 |
| FSRNet (Chen et al., 2018) | 4 × | 28.78 | 0.767 | 0.121 | 29.12 | 0.771 | 0.118 |
| HyFeat (Tomar et al., 2023c) | 4 × | 28.84 | 0.768 | 0.098 | 29.14 | 0.773 | 0.095 |
| SPARNet (Chen et al., 2020) | 4 × | 29.07 | 0.773 | 0.096 | 29.25 | 0.790 | 0.093 |
| SFMNet (Wang, Jiang, Zhong & Liu, 2023) | 4 × | 29.09 | 0.770 | 0.094 | 29.27 | 0.789 | 0.091 |
| Proposed | 4 × | **29.36** | **0.789** | **0.089** | **29.80** | **0.815** | **0.086** |
| SRCNN (Dong et al., 2014) | 8 × | 21.12 | 0.556 | 0.482 | 23.94 | 0.618 | 0.467 |
| VDSR (Kim et al., 2016) | 8 × | 21.47 | 0.568 | 0.464 | 24.12 | 0.636 | 0.462 |
| SRGAN (Ledig et al., 2017) | 8 × | 22.13 | 0.576 | 0.392 | 24.43 | 0.648 | 0.386 |
| URDGN (Yu & Porikli, 2016) | 8 × | 22.07 | 0.573 | 0.348 | 24.33 | 0.643 | 0.339 |
| SISN (Lu et al., 2021) | 8 × | 22.24 | 0.582 | 0.243 | 24.68 | 0.657 | 0.226 |
| FSRNet (Chen et al., 2018) | 8 × | 22.49 | 0.584 | 0.216 | 25.07 | 0.665 | 0.198 |
| DICNet (Ma et al., 2020) | 8 × | 22.64 | 0.587 | 0.192 | 25.19 | 0.668 | 0.188 |
| WIPA (Dastmalchi & Aghaeinia, 2022) | 8 × | 22.78 | 0.598 | 0.189 | 25.32 | 0.670 | 0.181 |
| HyFeat (Tomar et al., 2023c) | 8 × | 22.81 | 0.597 | 0.187 | 25.33 | 0.669 | 0.176 |
| SPARNet (Chen et al., 2020) | 8 × | 22.89 | 0.603 | 0.185 | 25.36 | 0.672 | 0.173 |
| SFMNet (Wang, Jiang, Zhong & Liu, 2023) | 8 × | 22.85 | 0.604 | 0.182 | 25.35 | 0.671 | 0.169 |
| CTCNet (Gao et al., 2023) | 8 × | 22.95 | 0.606 | 0.176 | 25.38 | 0.673 | 0.166 |
| Proposed | 8 × | **23.39** | **0.614** | **0.168** | **25.83** | **0.682** | **0.159** |




**Fig. 6.** Visual comparison of the reconstructed images with different SR methods for the scale of 4 × on CelebA-HQ, LFW dataset.

Figs. 4 and 5. A comprehensive comparison of recovered identities in the output faces is shown in Fig. 4, while Fig. 5 illustrates the specific or higher cosine similarity value carried by the fraction of images inferred in the test phase. It exhibits that the reconstructed images by the proposed framework are more credible and reasonably preserve the identity information than other existing face SR methods.

#### 4.3.2. Qualitative comparisons

The reconstructed faces through the proposed and different SR techniques as shown in Figs. 6 and 7 (see the regions corresponding to red box) are considered for qualitative performance comparison. A detailed comparative analysis based on these outcomes is presented below.

Bicubic is a baseline technique that is used to upsample the input images by computing the new intensity values from its neighboring pixels. Due to that, it expands the noise and generates poor-quality output images as shown in Figs. 6 and 7. The SRCNN (Dong et al., 2014) is a model that takes input with the same resolution as HR images. It uses a few convolutional layers for the task of estimating the HR images from the input. Although, the simplicity of the model makes the training process easier but it becomes hard to extract the deep features. Hence, the generated results (depicted in Figs. 6 and 7) fail to provide convincing facial details.

A deep structure-based VDSR model (Kim et al., 2016) recovers blurred images (as shown in fourth columns from left of Figs. 6 and 7) because of the gradient issue in the CNN layers which affects the learning ability of this model. The GAN-based frameworks such as SRGAN (Ledig et al., 2017) and URDGN (Yu & Porikli, 2016) use adversarial learning to restore edges but their training stability issues cause to restrict the parameter learning of these models. As a result, they reconstruct the output images with severe artifacts as exhibited in Figs. 6 and 7. An attention-based SISN SR method (Lu et al., 2021) utilized a feature split attention block to recover the landmarks of face images. This block splits the input features and enforces the model to focus on each of split regions. However, it may cause the model to put attention on those features which may not carry the beneficial details. Hence, the outcome of this as shown in Figs. 6 and 7 possess inappropriate contour of eyebrow and cheek. FSRNet (Chen et al., 2018) utilizes the estimated facial prior for the SR process and generates

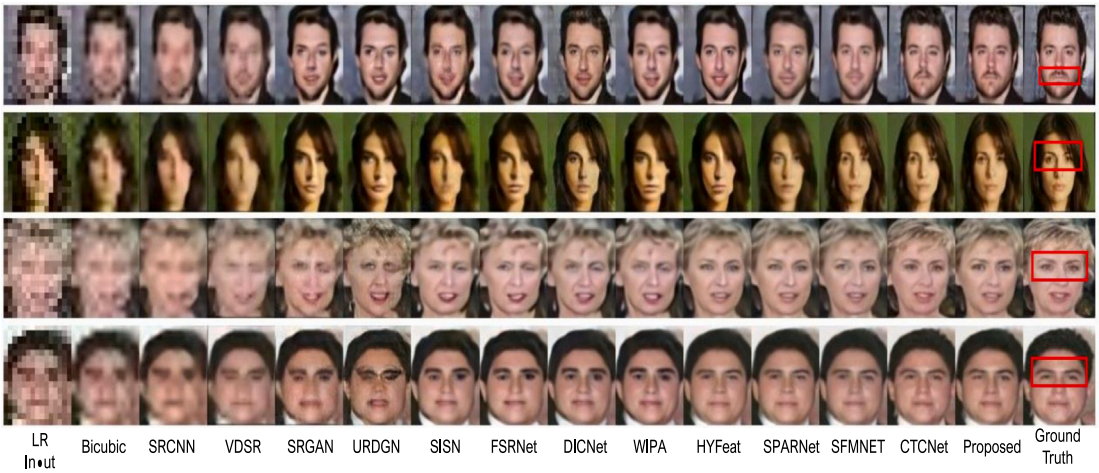



**Fig. 7.** Visual results comparison of the reconstructed images from different SR methods for scale of 8 × on CelebA-HQ, LFW dataset.

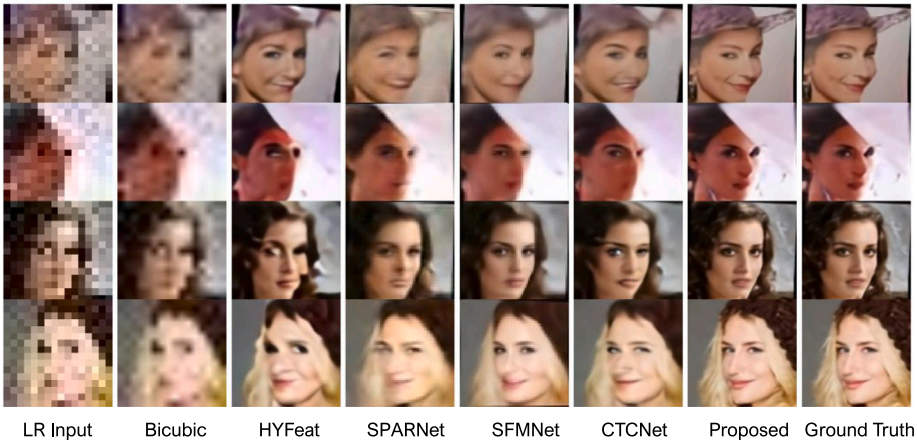



**Fig. 8.** Visual results comparison of different face SR methods under head pose variations.

visually clear face images as compared to other methods. However, it is hard to predict the accurate facial prior from the input images, and the use of inaccurate prior leads to reconstructing the images with distortion in key facial regions. An iterative collaboration-based approach DIC (Ma et al., 2020) iteratively performs the SR and then predicts the facial landmarks. The SR result of the previous iteration and the predicted prior are fed as input to refine the features iteratively. In some cases, the poor SR result may cause to propagate the inaccurate prior information iteratively and generate smooth face images with a lack of high-frequency details. The WIPA (Dastmalchi & Aghaeinia, 2022) integrates the wavelet estimation blocks with the CNN network to hold the high and low-frequency information. Although the predicted wavelet coefficients may be inaccurate due to that it fails to preserve the shape of eyes in the output faces. In Tomar et al. (2023c) the multi-scale features with attention in the attention block was utilized to learn better representation and focus on the valuable features that assist in preserving the texture details in the generated face images. However, the larger model size may cause it to overfit and degrade the SR performance. A novel attention-oriented method (Chen et al., 2020) has been developed to focus on features that carry valuable information which subsequently assists in recovering facial components. However, the generated HR images failed to properly reconstruct the eyes and nose due to ignorance of facial prior knowledge. An SR network has been proposed (Wang, Jiang, Zhong & Liu, 2023) that embeds the spatial branch and frequency branch to learn the local details and global details. After that, an interaction block was utilized to facilitate the interaction and fusion of spatial and frequency information to improve the model's representation capability. The complex interactions between the spatial and frequency branches may limit the model's generalization capabilities. The CTCNet (Gao et al., 2023) integrated the transformer with CNN to consider the local and global facial contexts during the SR process. It employs the feature cooperation module to maintain the consistency of the facial structure details. Further, the model fuses the multi-scale features to improve the feature representation ability. However, making the training balance of both components is highly tedious. Hence, generated images failed to preserve the realistic texture details as shown in Fig. 7. In the proposed framework, TN utilizes the FHGTP which guides the network to focus on carrying the details of facial components. Other than this, its output is also regularized through heatmap loss to maintain consistency in facial landmark space. Further, the SN adopts the distillation procedure to inherit the learned prior knowledge from the trained TN. In addition to that, regularization of the identity features takes place to constrain the identity in the resultant images. In this way, the proposed network effectively employs the facial prior with identity constraints for face super-resolution tasks. As a result, the proposed method demonstrates outstanding performance (refer Table 1) over the existing techniques in PSNR and SSIM scores. It can be seen from Figs. 6 and 7 that the proposed framework reconstructs face images with adequate structure of key facial components like eyes, nose, and cheek as compared to the previous SR methods. The proposed method recovers plausible textural features around the facial region compared to other methods under the head pose variations, as displayed in Fig. 8.

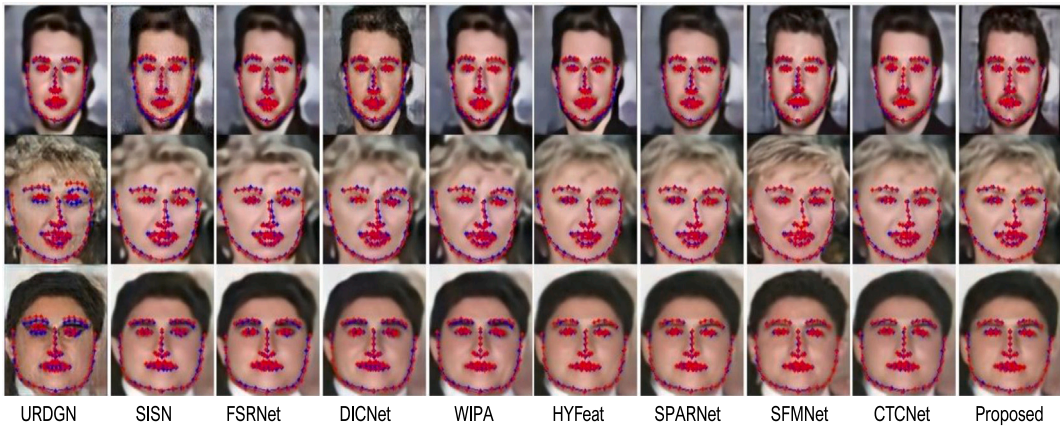



**Fig. 9.** Visual assessment of the reconstructed images by different SR approaches for holding the facial landmarks. The landmarks are denoted by blue and red dotted points corresponding to the ground truth HR and super-resolved images.

**Table 2**
Performance analysis of the proposed framework by using different combinations in the teacher network on the CelebA-HQ dataset for 4 × SR task.

| LR + FHGTP | w/o $\mathcal{L}_h$ | proposed | PSNR | SSIM | LPIPS | cosine similarity |
|---|---|---|---|---|---|---|
| ✓ | | | 29.15 | 0.772 | 0.093 | 0.61 |
| | ✓ | | 29.21 | 0.774 | 0.089 | 0.67 |
| | | ✓ | 29.80 | 0.815 | 0.086 | 0.71 |

**Table 3**
Formulation of different cases to determine the optimal value of hyperparameters.

| Case | $\lambda_{pix}$ | $\lambda_{id}$ | $\lambda_{sg}$ | $\lambda_f$ |
|---|---|---|---|---|
| 1 | 50 | 0.5 | 5 | 0.05 |
| 2 | 100 | 1 | 10 | 0.1 |
| 3 | 150 | 1.5 | 15 | 0.15 |

Although many existing methods (Chen et al., 2020; Dastmalchi & Aghaeinia, 2022; Lu et al., 2021; Ma et al., 2020) demonstrated considerable improvement in the visual perception of the output faces, but not accurate in maintaining the facial structure and landmarks particulars. Therefore, we evaluate the results of different SR methods for restoring facial landmarks. For this purpose the super-resolved images and counterpart Ground Truth images are provided as input to the face alignment model (Bulat & Tzimiropoulos, 2017) which estimates the landmarks of the face images (i.e., eyes, eyebrows, mouth, etc.). Finally, we visualize these predicted landmark attributes from the generated images of the different SR techniques to evaluate the consistency of preserving the facial structure details by them. The visual demonstration presented in Fig. 9 shows that the estimated landmark from the super-resolved images and counterpart ground truth images are completely overlapped in case of the proposed framework. It indicates that the super-resolved faces of the proposed framework carry more precise facial structural information than other compared SR methods.

Moreover, a few more experiments are conducted to verify the relevance of the proposed model for the real-world captured images. For this, we randomly selected the three face images, cropped the facial region, and preprocessed them by performing the downsample operation to make them compatible with the model's input. The obtained SR results for real-world face images are compared with a few SR methods, as shown in Fig. 10. It can be seen from Fig. 10 that the proposed model achieves more convincing visual results than other SR methods.

We also test the proposed model over the SCface dataset (Grgic, Delac, & Grgic, 2011), which contains the images captured by the surveillance camera. These images are preprocessed to make the input image with a size of 16 × 16 pixels, and then we perform the task of face SR. The visual comparison of obtained results from different SR methods are shown in Fig. 11. It can be clearly seen from the results that the proposed model generates the output face images with more facial details.

### 4.4. Analysis and discussion

This section examines the significance of the key components used in the proposed framework. It mainly analyzes the TN, SN, number of FAUs used, and computational complexity of the proposed framework.

**Analysis of the TN:** The TN provides the prior knowledge while reconstructing the HR images. Here, we experiment with the TN to analyze its impact on the performance of the proposed framework. The experimental results are reported in Table 2. Firstly, we removed the coarse SR network and replaced it with an input LR image. The TN takes the LR face image concatenated with FHGTP as input to train the framework. It achieves a PSNR and SSIM of 29.15 dB and 0.772 due to fewer details carried by the LR images than the coarse output image which causes a significant drop in the performance. Secondly, we train the TN without (w/o) $\mathcal{L}_h$ to analyze its effect on the outcome of the proposed framework. In this, the FHGTP is combined with coarse network output which contains better facial details than LR images. Hence, this combination helps the intermediate layers of the proposed framework to utilize the prior information better. However, their output is not regularized in the landmark space, which may cause them to learn the inappropriate landmark structure. It shows an improvement of 0.06 dB and 0.002 in PSNR and SSIM than the previous scenario but performs worse than the final setting in the proposed work.

**Analysis of the SN:** The combination of $L_{pix}$, $L_{id}$, $L_{sg}$, and $L_{Feature}$ is used to train the SN of the proposed framework. Here, we examine the significance of the components used in the loss function during the training of the SN. The corresponding results are presented in Fig. 12. Firstly, the $L_{pix}$, $L_{id}$, and $L_{sg}$ are used to train the SN which generates the images with poor facial landmark details, as shown in Column (b) of Fig. 12. It is due to the considerable gap between the layer's feature space of TN and SN which makes the learning procedure difficult for the SN. Then, we applied the $L_{pix}$, $L_{id}$, and $L_{Feature}$ which constrains the intermediate layer's features of the SN to adapt the FHGHP knowledge through the TN. The visual result depicted in Column (c) demonstrates the improvement in reconstructing eyes and lips details due to acquiring FHGHP knowledge. The visual result depicted in Column (d) indicates a better recovery of facial contour and texture details due to using the $L_{pix}$, $L_{sg}$, and $L_{Feature}$ which simultaneously regularizing the

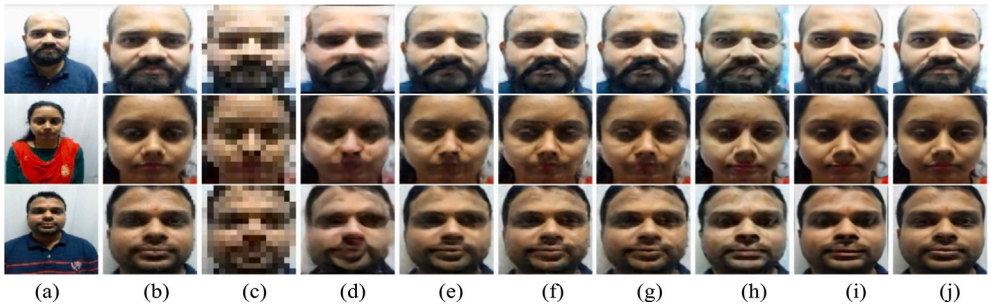



**Fig. 10.** SR results of the proposed framework and different existing models for real-world captured face images. (a) Real-world captured images, (b) cropped faces from (a), (c) LR version of (b), (d) DICNet, (e) WIPA, (f) HyFeat, (g) SPARNet, (h) SFMNet, (i) CTCNet, and (j) proposed framework.

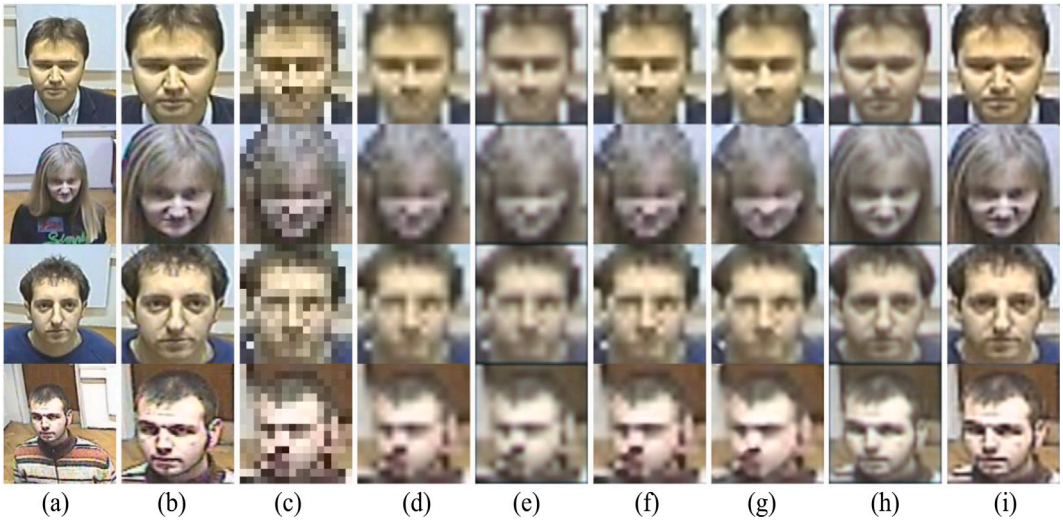



**Fig. 11.** Visual comparison of SR results on the captured images by the surveillance cameras (SCface dataset). (a) Captured images, (b) cropped faces from (a), (c) LR version of (b), (d) Bicubic, (e) HyFeat, (f) SPARNet, (g) SFMNet, (h) CTCNet, and (i) proposed framework.

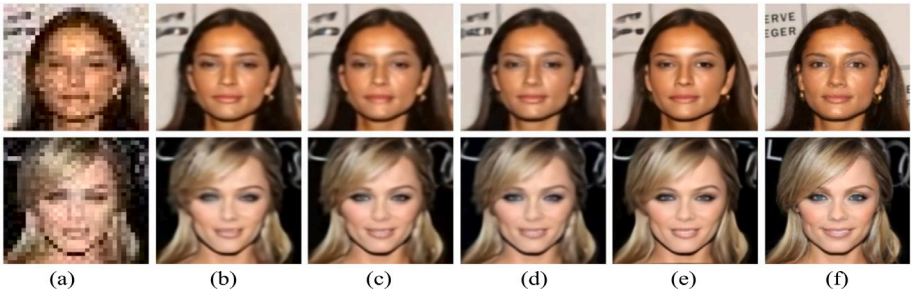



**Fig. 12.** Visual inspection of the reconstructed images using the different combinations in the loss function. (a) LR image, (b) using $L_{pix} + L_{id} + L_{sg}$, (c) using $L_{pix} + L_{id} + L_{Feature}$, (d) using $L_{pix} + L_{sg} + L_{Feature}$, (e) our proposed, and (f) ground truth HR image.

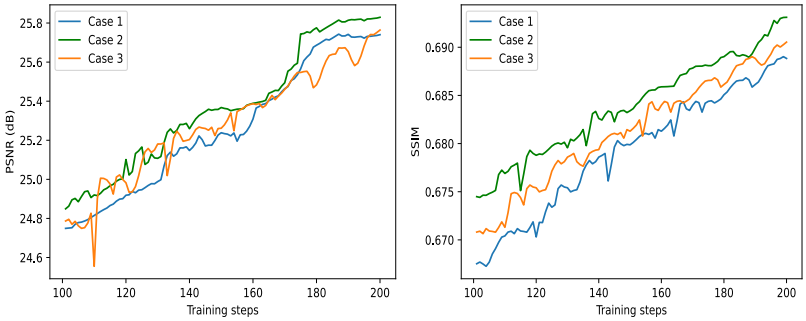



**Fig. 13.** Performance evaluation of the proposed framework under distinct values of hyperparameters.




**Fig. 14.** Ablation study to demonstrate the effectiveness of the proposed FAU. (a) LR image, (b) using only 3 × 3 filter with attention, (c) using only the 5 × 5 filter with attention, (d) first use the 3 × 3 filter then 5 × 5 filter followed by attention, (e) proposed FAU, (f) SR result of the proposed framework, and (g) ground truth HR image.

**Table 4**
Performance analysis of the proposed framework under different combinations of varying size filters in the proposed FAU.

| 3 × 3 | 5 × 5 | 3 × 3 → 5 × 5 | Proposed | PSNR | SSIM | LPIPS | Cosine similarity |
|---|---|---|---|---|---|---|---|
| ✓ | | | | 24.87 | 0.642 | 0.187 | 0.59 |
| | ✓ | | | 25.12 | 0.657 | 0.179 | 0.61 |
| | | ✓ | | 25.29 | 0.674 | 0.171 | 0.64 |
| | | | ✓ | 25.83 | 0.682 | 0.159 | 0.68 |

layer's feature space and output of SN. It enables the SN to utilize the FHGTP knowledge efficiently while generating HR images.

**Hyperparameter Tuning**: The hyperparameter tuning is performed to balance the multiple terms used in Eq. (11). It handles the degree by which these terms contribute to the model output. For this, we formulate the cases with different settings which are shown in Table 3. The proposed model is trained with an assemble loss function using a distinct set of values and the corresponding results are demonstrated in Fig. 13. For small values of $\lambda_{pix}$, $\lambda_{id}$, $\lambda_f$, $\lambda_{sg}$ the model is unable to restrict the output properly in different spaces, while higher values of these hyperparameters may lead to prioritize the low-frequency content over high-frequent details. Hence, it can be seen from Fig. 13 that the model achieves better performance under $\lambda_{pix} = 100$, $\lambda_{id} = 1$, $\lambda_f = 0.1$, and $\lambda_{sg} = 10$, respectively.

**Effectiveness of FAU**: We conducted the ablation study to demonstrate the effectiveness of the FAU designed in the proposed model. The intermediate feature maps corresponding to these experiments are shown in Fig. 14 and their objective assessment is displayed in Table 4. In this direction, we modify the FAU by using the different filters and then keep them in different orientations. Firstly, the 3 × 3 filter is used alone with the attention unit, making it hard to learn the various pattern features, as shown in Fig. 14(b). This may be due to their relatively small receptive field. Secondly, the 5 × 5 filter is applied instead of 3 × 3 in the previous case which slightly considers broader context but may not be able to capture the long-range dependencies as the results depicted in Column (c) of Fig. 14. After that, both the filters are placed one after the other sequentially, followed by the attention unit. It may perform better than the above scenarios because it learns coarse and fine details from the different scale filters. However, This may limit the model's ability, as seen from Column (d) of Fig. 14 to capture global context because of the dependency on the outcome of the previous filter. The results of the final setting of FAU are shown in Column (e) and Column (f) of Fig. 14 showing better recovery of features than other configurations. It utilizes the two branches to process the input independently at different scales which assists in capturing multi-scale features. These features help the proposed model learn better representation compared to the above-discussed scenarios.

**Number of FAUs:** The proposed framework repeats the FAUs to enhance the feature representation capability. The increasing number of FAUs improves performance to a certain extent, and then overfitting problems may occur due to a massive number of parameters. The experimental results are demonstrated in Fig. 15 and Table 5 with different numbers of FAUs. We set the 9 FAUs in the proposed framework based on the experimental results demonstrated in Fig. 15 and Table 5.

**Computational complexity:** Table 6 shows the comparison of models in terms of parameters, running time, and number of floating point operations (FLOPs) involved in performing the SR task. The proposed framework contains fewer parameters and requires a shorter running time than most SR methods to generate HR images. Although the proposed framework performs higher FLOPs, but it demonstrates better performance in terms of PSNR, SSIM, identity similarity, and visual appearance of SR results.

**Table 5**
Performance analysis of the proposed method under different number of FAUs.

| No. of FAUs | 7 | 8 | 9 | 10 |
|---|---|---|---|---|
| PSNR | 25.41 | 25.48 | 25.83 | 25.62 |
| SSIM | 0.667 | 0.672 | 0.682 | 0.674 |

**Table 6**
Complexity comparison of different SR methods.

| SR method | parameters (M) | running time (ms) | FLOPs (G) |
|---|---|---|---|
| URDGN (Yu & Porikli, 2016) | 1 | 45 | 14.6 |
| FSRNet (Chen et al., 2018) | 3.1 | 53 | 39 |
| DICNet (Ma et al., 2020) | 20.8 | 65 | 14.8 |
| SPARNet (Chen et al., 2020) | 10 | 51 | 7.2 |
| CTCNet (Gao et al., 2023) | 24.4 | 95 | 47.2 |
| Proposed | 5.3 | 49 | 35 |

## 5. Conclusions

This paper presented a distillation-oriented teacher and student network for reconstructing the high-quality output face images from the noisy low-resolution face images captured through low-quality consumer imaging devices. The proposed framework embeds the feature attention unit which uses multi-scale information to represent the input more reasonably. Especially, the teacher network exploits the facial heatmap-based ground truth prior that directs the intermediate layers to discover the structure-specific features. Then, the student network regularizes the identity features and utilizes the learned prior knowledge through the distillation procedure to obtain the super-resolved result. It helps to maintain the consistency of identity information and facial structure details in the output images. The extensive experiments and comparison of output images established indicate the superiority of the proposed framework over the existing techniques for restoring essential details in the output images. The outcome of this research implies the effectiveness of facial structure information and identity constraints to improve SR performance.

This research work can be extended in the future by combining the multiple attention mechanisms to improve the face SR results. Moreover, the robust face SR methods could be developed to handle multiple degradations.

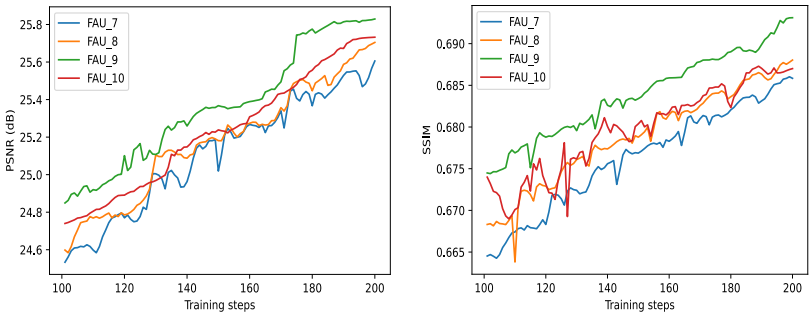



**Fig. 15.** Effectiveness of different numbers of FAUs on the performance of the proposed framework.

## CRediT authorship contribution statement

**Anurag Singh Tomar:** Conceptualization, Methodology, Software, Writing – original draft, Visualization, Investigation, Writing. **K.V. Arya:** Supervision, Reviewing, Validation. **Shyam Singh Rajput:** Conceptualization, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

No data was used for the research described in the article.

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